

Project Proposal

NestNav: AI-Powered Apartment Finder Virtual Assistant

Latha Boralingaiah(018301361)

1. Use Case Description

NestNav is an LLM-based virtual assistant designed to assist users find rental housing and. The virtual assistant tackles the challenge of navigating thousands of listings across multiple platforms by providing customized recommendations, price comparisons, neighborhood analysis, and real-time market insights.

Target Users: Students seeking housing near campus, young professionals relocating for work, budget-conscious renters, and anyone searching for apartments.

Why This Use Case: Every query requires considerable tool use (listings/neighborhoods search using database or looking up current prices/trends on web), making it ideal for demonstrating tool-calling capabilities of an LLM. This helps to solve a real problem currently faced by students and professionals.

2. Example User Queries

#	User Query	#	User Query
1	Find 2-bedroom apartments in Sacramento for under \$2500	11	What amenities are common in this area?
2	What's the average rent for 1BR in Jacksonville?	12	Is rent going up or down in Columbus?
3	Compare rent prices: San Jose vs Sacramento	13	Find apartments near bus/train stations
4	Pet-friendly apartments in Sacramento	14	What's typically included in rent?
5	Is this neighborhood safe?	15	Best neighborhoods for young professionals
6	Apartments with in-unit laundry	16	Find month-to-month lease options
7	What's the commute time to downtown Columbus?	17	What questions should I ask the landlord?
8	Show me safest neighborhoods under \$2000	18	Calculate total monthly cost with utilities
9	Apartments with parking included	19	Studios vs 1-bedroom price difference?
10	Compare walk scores of these two areas	20	Create apartment checklist for my budget

3. Dataset

Primary Dataset: USA Housing Listings (Craigslist) - Kaggle

Link: <https://www.kaggle.com/datasets/austinreese/usa-housing-listings>

4. LLM Selection

Model	Params	Role	Justification
Phi-3.5-mini-instruct	3.8B	Smaller	Microsoft's compact model, excellent instruction-following
Mistral-7B-Instruct-v0.3	7B	Larger	State-of-the-art 7B model, 4-bit quantization fits Colab